

Modern Technologies Transforming the Future

1. Introduction + Quantum Computing

- Meaning of modern tech
- Quantum concepts: Qubits, Superposition, Entanglement
- Applications: Cryptography, Pharma, Finance

2. Neural Networks

- Structure: Input, Hidden, Output
- Types: ANN, CNN, RNN
- Uses: Vision, Recommendation, Fraud detection

3. Deep Learning

- Many layered neural networks
- CNNs, RNNs, Transformers
- Applications: Medical imaging, Self-driving cars, Generative AI

4. Natural Language Processing

- Language understanding by machines
- Tokenization, Sentiment, NER
- LLMs: BERT, GPT
- Applications: Chatbots, Search, Assistants

5. Game Development

- Design, Programming, Graphics
- Unity, Unreal Engine
- AI & NLP in modern games

Conclusion

These technologies are shaping the future across industries.